

Microbiological Techniques

Laboratory Report — Serial Dilution, Spread Plate & Pour Plate

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Technique category: Quantitative Microbiology

1. Objective

To determine the bacterial concentration (CFU/mL) of a culture sample using serial dilution followed by plating, and to calculate colony-forming units from colony counts within the statistically countable range.

2. Introduction

Serial dilution reduces a bacterial suspension to a countable concentration before plating, since undiluted samples typically produce confluent, uncountable growth. Spread plating distributes a known volume of diluted sample across the agar surface, producing surface colonies; pour plating mixes the sample into molten agar before it sets, producing colonies throughout the agar depth. Counting colonies on a plate within the countable range (30–300 colonies) and applying the dilution factor and plated volume gives the original sample's bacterial concentration in colony-forming units per mL (CFU/mL).

3. Materials

- Bacterial culture/sample and sterile diluent (saline or phosphate buffer)
- Sterile pipettes/micropipettes and tips
- Sterile Petri dishes and molten nutrient agar (pour plate) or pre-poured plates (spread plate)
- Sterile spreader (spread plate)
- Incubator (35–37°C) and colony counter

4. Method

- Performed serial 1:10 dilutions from the original sample across a dilution range (10^{-2} to 10^{-5}).
- Plated a known volume of each dilution by spread plate and/or pour plate method.
- Incubated plates inverted at 35–37°C for 18–24 hours.
- Counted colonies on each plate, noting which fell within the countable range (30–300).
- Calculated CFU/mL for each countable plate and averaged across valid replicates.

5. Results

Four dilutions (10^{-2} through 10^{-5}) were plated. The 10^{-2} plate was too numerous to count (TNTC), and the 10^{-5} plate yielded only 4 colonies — both outside the countable range and excluded from calculation. The two remaining plates (10^{-3} and 10^{-4}) fell within the countable range.

Dilution	Volume (mL)	Colonies	Countable?	CFU/mL
10^{-2}	0.1	TNTC	No	—

10 ⁻³	0.1	212	Yes	2.12 × 10 ⁶
10 ⁻⁴	0.1	34	Yes	3.40 × 10 ⁶
10 ⁻⁵	0.1	4	No	—

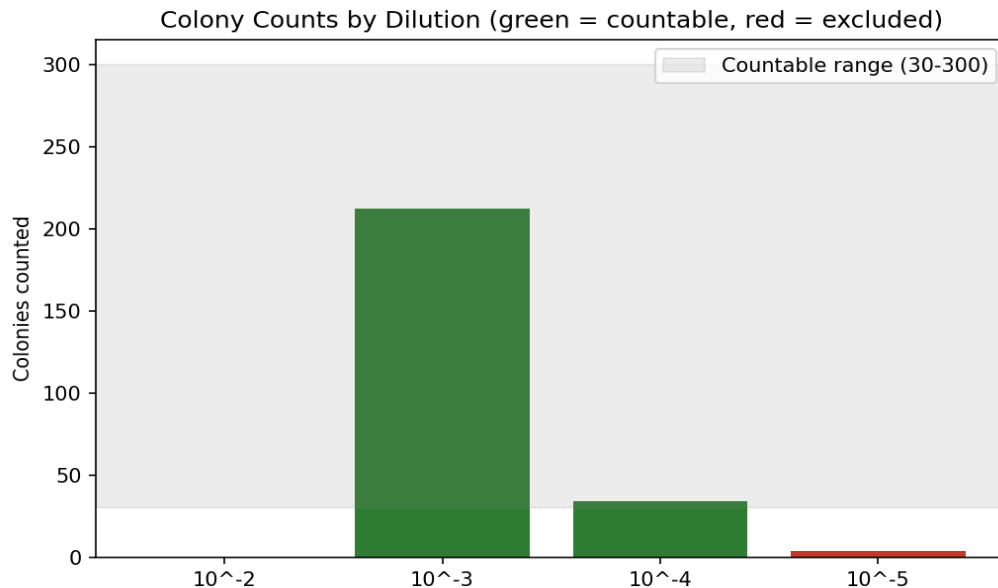


Figure 1. Colony counts by dilution, with the countable range (30–300) shaded; green bars mark plates used in the final calculation.

Averaging across the two countable plates gave a final estimated concentration of approximately **2.76 × 10⁶ CFU/mL** in the original sample.

6. Discussion

The two countable plates gave reasonably consistent CFU/mL estimates (2.12×10^6 and 3.40×10^6), a roughly 1.6-fold difference that falls within the range commonly seen between adjacent dilutions due to normal pipetting and plating variability. Using only countable plates for the final estimate, rather than including the TNTC or very-low-count plates, is important because counts outside the 30–300 range are statistically less reliable — overcrowded plates suffer from overlapping colonies that undercount true density, while very low counts carry high relative statistical error from a small sample size.

7. Conclusion

This exercise demonstrated the standard quantitative microbiology workflow of serial dilution and plating, and reinforced why only plates within the statistically countable range should be used when calculating final CFU/mL estimates.